

Chapter 48 — Lower Urinary Tract Reconstruction in Children

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- **Functions a reconstructed lower tract must achieve:** protect renal function → avoid urinary tract infection (UTI) → eventually provide urinary continence. Operative intervention is minimised, but these functions are non-negotiable.
- **Thorough individual evaluation before reconstruction:** each child's pathophysiology is understood so that techniques are applied thoughtfully, optimising results and minimising morbidity.
- **Scope:** augmentation cystoplasty (intestinal segments used more extensively here than anywhere else in urology), procedures that raise **bladder neck and external sphincter resistance**, and **catheterisable channels**.
- **Clean intermittent catheterisation (CIC)** — Lapedes et al. (1972, 1976), built on Guttman and Frankel (1966) — is the **single most important contribution to lower tract reconstruction**. Many children do not void adequately after reconstruction, so a **reliable, easy means of catheterisation** is often essential.
- **Primary indication:** correction of **bladder "hostility" unresponsive to medical management**.
- **Spina bifida** accounts for many patients; results in the chapter therefore focus on the **neurogenic population**.
- **Most important determinant of outcome:** **commitment of the patient and family** to good care — difficult to gauge, never to be underestimated.

"FUNCTIONAL" URINARY TRACT

- **Two dynamic phases**, stated as a pair: **passive storage** — reservoir holding urine at low pressure without leakage; **active voiding** — contraction with efficient elimination.

Dysfunction

Upper Urinary Tract

- **Understand the dynamics of the entire urinary tract** before any major reconstruction.
- **Hydronephrosis → exclude upper tract obstruction;** obstruction may itself be secondary to severe, long-standing hostility (poor compliance and poor emptying).
- **Nuclear renography with a urethral catheter** in the bladder rules out primary upper tract obstruction.
- **Obstruction, if present, is corrected at the time of bladder and sphincter reconstruction.**
- **Vesicoureteral reflux (VUR) with hostility** may be primary or secondary and is hard to distinguish; **new-onset reflux absent on earlier evaluation is likely secondary;** in neurogenic children reflux is **often secondary**.
- **Secondary reflux often resolves with augmentation alone** once the bladder is adequately managed; whether reflux matters at all in a large, compliant bladder is debated.
- **Bacteria ascend with or without reflux** after reconstruction; some continent diversions without any antireflux mechanism show **no increase in pyelonephritis when compliance is adequate**.
- **Authors' practice:** expect most secondary reflux to resolve, but **correct high-grade reflux** at reconstruction.
- **Upper tract dysfunction presents as hydronephrosis, pyelonephritis or impaired renal function;** all are addressed before and at reconstruction.

Bladder Dysfunction

- **Assess each component of passive and active function independently.**
- **Pathogenic threshold: filling pressure >40 cm H₂O reached chronically** (McGuire et al., 1981) → impaired ureteral drainage → pyelocalyceal change, hydroureteronephrosis, falling glomerular filtration rate (GFR). Persistent elevation also causes **acquired VUR**.
- **Pharmacotherapy plus intermittent catheterisation** lowers filling pressure, particularly with overactive detrusor contractions, and is effective in neurogenic children.
- **Compliance unresponsive to medical management → augmentation cystoplasty.**
- **After reconstruction, efficient detrusor contraction and emptying become less likely → CIC is taught and accepted preoperatively.** CIC is what permits aggressive correction of storage; spontaneous voiding is a goal, not a requirement.
- **Continence requires outflow resistance (bladder neck + external sphincter) greater than bladder pressure** throughout daily activity; a deficient outlet leaks.
- **α-adrenergic agents** can raise outflow resistance, but **operative reconstruction is more commonly required.**
- **Filling-phase incontinence from a poor outlet: evaluate the detrusor as well.** Once outlet resistance is raised operatively, **adverse detrusor behaviour may be unmasked** — high-pressure storage or uninhibited contractions not previously documented → **provocative urodynamics with bladder neck occlusion** before any bladder neck reconstruction.
- **Normal synergic voiding:** bladder neck descends, relaxes and opens → external sphincter relaxes → detrusor contracts → low-pressure voiding.
- **Dyssynergia** (uncoordinated bladder neck, sphincter and detrusor) → chronic high-pressure voiding that damages bladder and upper tracts; a **fixed, fibrotic external sphincter** behaves the same way. **Initial treatment: pharmacotherapy and CIC** to bypass the abnormal voiding mechanics.

Other Considerations

- **Assess renal function in every patient**, particularly with hydronephrosis or severe scarring.
- **Urinary solutes, particularly chloride, are absorbed by small- and large-bowel mucosa.**
- **Normal kidneys handle the reabsorbed chloride and acid load; reduced renal function → significant metabolic acidosis.** Pre-existing acidosis invariably worsens when urine is stored in small or large bowel.
- **Concentrating ability is the first renal function lost** after obstruction or infection → very large urine volumes; the **reconstructed capacity must hold output for 3–4 hours.**
- **Oliguria** (renal failure) → greater collection and inspissation of mucus in the augmented bladder.
- **Hepatic failure: ammonia reabsorption is dangerous;** drugs excreted in urine may be reabsorbed → **liver function tests and arterial blood gases** where appropriate.
- **Take a careful bowel history: short gut** is a concern after cloacal exstrophy, bowel resection or significant radiation; **chronic diarrhoea or faecal incontinence → caution before using the ileocaecal valve.**
- **Commitment: incontinence can protect** some patients from infection and upper tract deterioration; **effective storage without regular emptying is dangerous.** Everyone must understand the responsibility that continence brings.
- **Timing: early** when upper tracts or renal function are threatened (high outflow resistance with poor compliance) — augmentation may slow decline even with established insufficiency, but **prevention is preferable; more commonly later, for continence**, when incontinence or a temporary stoma becomes socially unacceptable (varies by family).

- **Identify all needs and address them in one operation** where possible (not always feasible when intervening for infection or hydronephrosis): urodynamics decide the need for an outlet procedure; **preoperative CIC is mandatory** — it demonstrates commitment and shows whether a **continent abdominal wall stoma** is needed for reliability and independence; identify neurogenic children who would benefit from an **antegrade colonic enema** for constipation. Sequential procedures add morbidity and technical difficulty.

Concept — every reconstructive decision is a **storage-emptying trade**: raising capacity or outlet resistance converts a leaking but self-venting system into one that depends entirely on reliable emptying, which is why CIC and commitment are prerequisites rather than afterthoughts.

PATIENT EVALUATION

- **Upper tract imaging in every patient** before reconstruction; most already have surveillance ultrasonography.
- **Hydronephrosis** → **assess for obstruction and reflux**: nuclear renography with a bladder catheter for primary obstruction; **voiding cystourethrogram (VCUG) or video urodynamics** for reflux.
- **With hydronephrosis**: **baseline serum electrolytes, blood urea nitrogen (BUN) and creatinine**; raised creatinine or significant hydronephrosis → **nephrology evaluation and 24-hour urine** for creatinine clearance and volume.
- **Prior diversion**: permanent intestinal diversion in children is now largely confined to **cystectomy for cancer**; most diversions are temporary. **Key to undiversion**: **understand the original pathology** that led to diversion.

Urodynamics

Bladder Dynamics: Capacity and Compliance

- **Urodynamic testing is essential**; reproducible in children only with meticulous technique, and **interrater reliability of urodynamic study (UDS) interpretation is variable** even across experienced centres.
- **Urologic Management to Preserve Initial Renal function (UMPIRE) protocol**: iterative protocol for best practice in paediatric neurogenic bladder; its standardisation efforts **improved interpretive consensus**.
- **Artefact sources**:
 - **Transurethral catheter size** alters leak point pressure, voiding pressure and emptying, especially in infants and young males; a **suprapubic catheter avoids this** but is usually impractical.
 - **Medium**: body-temperature saline or iodinated contrast gives reproducible results.
 - **Filling rate** can dramatically change end filling pressure and therefore compliance → **fill at ≤10% of predicted capacity per minute**.
- **Interpretation for planning**:
 - **Cystometric capacity + fluoroscopic bladder appearance** → length of bowel segment.
 - **Detrusor leak point pressure + end filling pressure + fluoroscopic bladder neck** → need for a bladder neck procedure.
 - **VUR + history of UTI** → ureteral reimplantation at reconstruction.

Sphincter Dynamics: Outflow Resistance

- **Bladder neck and external sphincter act synergically, but only one is required for continence.**
- **Neurogenic dysfunction** → diminished storage resistance and/or dyssynergic voiding.

- **Electromyography (EMG) of the external sphincter** is required to assess coordination and dyssynergia: perineal surface electrodes, abdominal sensors, anal plugs, vaginal monitors, wire and concentric needle electrodes.
- **Concentric needle or dual wire electrodes through a 25-gauge needle are more accurate** in neurogenic children; UMPIRE data **question the reliability of surface electrodes**; few centres offer needle studies; the role of sphincter analysis **remains unclear**.

Bladder Emptying

- **Parameters:** synergic sphincter relaxation on EMG, flow rate, postvoid residual.
- **Neurologically intact good emptiers are much more likely to keep emptying after reconstruction** than neurogenic patients.
- **No test ensures spontaneous voiding after augmentation → every patient must be prepared for CIC.**
- **Examine the native urethra for ease of catheterisation;** CIC is learnt and practised preoperatively until patient, family and surgeon are confident; physical and psychosocial limitations affect independent care.
- **Failure to catheterise reliably → upper tract deterioration, UTI or bladder perforation — despite a technically perfect operation.**
- **Given the choice, patients overwhelmingly prefer an abdominal wall stoma** to the native urethra.

Concept — *urodynamics predicts storage but not emptying: capacity, compliance and leak pressures can be measured and planned for, whereas postoperative voiding cannot, so every plan assumes CIC.*

PATIENT PREPARATION

- **Optimise general status:** nutrition and hydration corrected; cardiac and pulmonary comorbidity well managed.

Preoperative Considerations

- **Bowel preparation has been challenged by enhanced recovery after surgery (ERAS) protocols,** adopted only recently in neurogenic lower tract surgery; paediatric surgical evidence shows **omitting bowel preparation does not increase complications or surgical site infection.**
- **Hydration is critical** (long operations, large fluid shifts) — intravenous fluids on the morning of surgery if needed.
- **Oral antibiotic bowel preparation: once dogma, now personal preference.**
- **ERAS elements: clear-liquid carbohydrate loading up to 2 hours preoperatively;** multi-day clear diet and formal mechanical preparation abandoned; ERAS associated with **lower 90-day complications** and **earlier bowel function and diet** after augmentation.
- **Barriers to ERAS standardisation:** clinician unfamiliarity, resources and administrative support, lack of consensus.
- **Urine culture several days preoperatively;** positive → treat (oral or intravesical antibiotics) and **document a sterile repeat culture.**
- **Ventriculoperitoneal (VP) shunt** (common in spina bifida): with sterile urine and good bowel preparation, **shunt infection risk should be very low;** however, a Pediatric Health Information System (PHIS) database analysis showed **increased shunt failure after augmentation,** cause unclear and likely multifactorial.

KEY POINTS: EVALUATION AND PREPARATION

- Evaluate renal function, upper tract obstruction and VUR if hydronephrosis is present, although most such changes are secondary to bladder dysfunction.

- Careful urodynamic evaluation is critical to understand every problem contributing to dysfunction.
- The patient will likely require CIC after reconstruction and must be capable of performing it reliably.
- Commitment of patient and family is essential.
- Bowel preparation is surgeon preference, but sterilising the urine has its advantages.
- ERAS protocols show promise but need multicentre prospective studies to become standard practice.

Concept — preparation has **shifted from the bowel to the urine**: mechanical bowel preparation is now optional under ERAS, whereas sterile urine remains mandatory.

ANTIREFLUX

- **Reimplantation into the native bladder**, dilated ureter or not, is standard, with good long-term success and rare complications — **preferable and usually possible** during reconstruction.
- **Chronically scarred, short ureters after diversion** need special attention; in Hendren's undiversion experience **most complications arose from those ureters**.

Transureteroureterostomy (Figure 48.1, Hendren)

- **Donor ureter crosses the retroperitoneum without tension**, sometimes **above the inferior mesenteric artery**; **avoid angulation beneath the inferior mesenteric vessels**; anastomosis with **interrupted sutures, knots outside**.

Psoas Hitch

- **Purpose**: precise control of reimplant length and direction; **prevents ureteral angulation with bladder filling** — particularly obstructive with a dilated, scarred ureter.
- **Nonabsorbable monofilament sutures to psoas muscle and fascia**, never entering the bladder lumen — otherwise stones form.
- **Broad, shallow purchases** hold long term; **avoid trapping the femoral and genitofemoral nerves**; **do not tie so tightly that sutures cut through** bladder or psoas.
- **Divide the contralateral bladder pedicle** when extra mobility and hitch length are needed.
- **Order**: Hendren and Hensle (1980) reimplant before hitching; **do whichever order allows each step precisely with good visualisation**. Full technique in Chapter 124.

Technique (Figure 48.2)

Traction sutures; new hiatus **as high as possible** → two retractors stretch the back wall taut → incise mucosa and lay back flaps (in a good bladder, a submucosal tunnel without flaps) → ureter **widely mobilised with surrounding tissue, often with gonadal vessels** → **long-tunnel reimplant (8–10 cm)** when the ureter is abnormal or a tapered bowel segment is implanted → two fingers tent the bladder to the psoas, **as high as possible and lateral to the iliac vessels** → **3–5 sutures through bladder muscle, not mucosa**, big deep bites of psoas including tendon, avoiding nerves → **tie only after all are placed**; **surgeon holds bladder against psoas while assistant ties** → hitch lies just lateral to, and immobilises, the new hiatus; bladder elongated and asymmetric, allowing a very long tunnel.

Antireflux With Intestinal Segments

- **The need to reimplant into bowel may determine the segment chosen** for augmentation or replacement.
- **Colon**: from ureterosigmoidostomy and colon conduit experience — **flap valve by tunnelling the ureter beneath a taenia**; principles: **direct mucosa-to-mucosa anastomosis + submucosal tunnel of adequate length**; durable since the 1950s (Nesbit, 1949; Goodwin et al., 1953; Leadbetter and Clarke, 1954), based on **Coffey (1911)**.

- **Implant from within the reservoir or from outside** after the segment is reconfigured and closed.
- **Stomach** (rare): reimplant as in native bladder, same tunnel length-to-ureteral width principles.
- **Ileum: effective antireflux is most difficult** — requires nipple valve techniques.

ANTIREFLUX OPTIONS BY RECEIVING TISSUE

Tissue	Mechanism	Difficulty
Native bladder	Submucosal tunnel ($\pm$ psoas hitch)	Preferred ; standard
Colon	Tunnel beneath a taenia ; mucosa-to-mucosa	Established, durable
Stomach	Submucosal tunnel as in bladder	Feasible; rarely needed
Ileum	Nipple valve (Kock / hemi-Kock)	Most difficult

- **Pattern:** tissues with a supporting muscular wall accept a flap-valve tunnel; ileum does not, and needs a nipple. **Trap:** assuming antireflux is equally easy in any bowel segment.

KEY POINTS: ANTIREFLUX

- Ureteral reimplantation into native bladder is preferable when necessary; the initial bladder incision may facilitate it.
- An antireflux mechanism can be created with any bowel segment but is most challenging with ileum.

Concept — *antireflux is a **flap-valve problem**: any tissue with a supported submucosa — bladder, colon under a taenia, stomach — accepts a tunnel of adequate length for the ureteral width, whereas ileum cannot and needs a nipple.*

BLADDER NECK RECONSTRUCTION

- **Reliable outflow resistance is one of the greatest technical challenges** of bladder reconstruction.
- **Cause neurogenic or anatomic:** exstrophy, bilateral ectopic ureters, ureteroceles.
- **Many techniques exist, so no single option is best;** thorough knowledge of the patient's physiology is required.
- **Technique selection depends on** whether a sustained contraction can empty the bladder and whether hyperreflexic contractions are absent.
- **Every outlet procedure raises resistance at the expense of detrusor contractility.**
- **Raising outlet resistance can create a noncompliant, hostile detrusor** for unclear reasons; **provocative cystometry uncovers only some patients at risk → close postoperative observation is mandatory.**
- **Options:** flap valve mechanisms, artificial or autologous bulking agents, slings, artificial urinary sphincter (AUS) — individualised to pathology, needs, goals and desire for independence.
- **Many repairs prohibit spontaneous voiding;** neurogenic children needing augmentation plus an outlet procedure need CIC; other diagnoses without augmentation may void.
- **Results derive mostly from spinal dysraphism;** every technique has a **learning curve** requiring forthright reporting.
- **Evidence is limited:** no consistent definition of "success" or "continence", mixing of patients with and without augmentation, small mixed populations.

BLADDER NECK AND OUTLET PROCEDURES COMPARED

Procedure	Mechanism	Voiding	Dominant problem	Place
Young-Dees-Leadbetter	Tubularised trigone, lengthened urethra	Possible (exstrophy)	Poor tone of neurogenic muscle	Exstrophy; largely abandoned in neurogenic
Fascial sling / wrap	Coaptation + elevation	Rarely adequate → CIC	Overcome by poor compliance	Better in females ; with augmentation
Bladder neck injection	Subepithelial bulking, coaptation	Preserved	Poor durability	Limited, select group
AUS	Cuff compression, cyclable	Preserved	Erosion, infection, malfunction	Patients who void
Kropp	Flap valve (tube in trigonal tunnel)	Abolished	Difficult catheterisation ; no pop-off	High continence
Pippi Salle	Flap valve (anterior onlay)	Abolished	Fistula; lower continence	Easier catheterisation
Bladder neck division	Complete separation	Abolished; stoma obligatory	No urethral access or pop-off	After failed repairs

- **Pattern:** the more complete the continence mechanism, the more completely voiding and pop-off are lost; only the AUS preserves voiding. **Trap:** treating any neurogenic outlet repair as complete without planned augmentation and CIC.

Young-Dees-Leadbetter Repair

- **One of the most recognised outlet procedures;** remains a **primary consideration for the exstrophic bladder neck.**

Technique

Young (1919): excise part of the bladder neck and tighten it over a silver probe → **Dees (1949):** extend the excision through the trigone → **Leadbetter (1964):** elevate the ureters to a cephalad position on the bladder floor → **tubularised trigone and lengthened urethra.** Full description in Chapter 43.

Results

- **Minimal success in neurogenic dysfunction,** attributed to **poor tone and activity of the wrapped neurogenic muscle;** many early patients were not augmented.
- **Contrast with exstrophy:** neurogenic patients mostly need **augmentation and CIC,** and continence often needed a **second operation.**
- **Largely fallen out of favour in neurogenic dysfunction.**
- **Silicone sheath** (Mitchell and Rink, 1983): external support and coaptation, plus a plane for a future AUS cuff → **thick Silastic sheaths eroded** into bladder or urethra; **thinner sheath with interposed omentum:** less erosion, better continence, easier later cuff.
- **Largest neurogenic series:** continence achievable, but **at the cost of near-universal augmentation and multiple procedures;** sheathed patients initially continent, then erosion; mirrored by other series.

Fascial Sling

- **Mechanism: coaptation and elevation of the bladder neck,** raising resistance above resting pressure and the pressure of stress or Valsalva.
- **The bladder neck is fixed:** a strong contraction may produce flow but **rarely adequate emptying** in anatomic or neurologic disease → **prepare for CIC.**
- **Sling resistance is overcome by overactivity or poor compliance** → **simultaneous augmentation** is reported in most children who become continent.

- **Snodgrass exception:** good results without augmentation, but children with **sustained pressures >40 cm H₂O on anticholinergics** developed reflux or hydronephrosis; long-term follow-up of the same cohort showed frequent further continence procedures, augmentation, upper tract change and chronic kidney disease (CKD) — **a hostile bladder after raising outlet resistance is a major risk.**
- **Isolated outlet procedures in neurogenic patients:** a substantial proportion progress to augmentation or onabotulinumtoxinA injection, **most within 2 years** and within 5 years → **surveillance for a minimum of 5 years.**
- **Expanded polytetrafluoroethylene (Gore-Tex):** early continence not maintained; **biodegradable scaffolds:** good early results.

Technique

Clear fat over the bladder neck and lateral endopelvic fascia → incise endopelvic fascia for about **2 cm** → locate the bladder neck–urethral junction by **gentle traction lodging the catheter balloon** → blunt plane between posterior bladder neck and **vagina (female) or rectum (male)**; often easier **from the cul-de-sac**, behind bladder and ureters from above → obscured landmarks (secondary repair): **open the bladder** to avoid entering urethra or posterior structures; a **transvaginal approach** avoids both.

- **Fascia:** based on **McGuire and Lytton (1978)**; **rectus fascia 1 cm wide**, harvested vertically or horizontally per incision; other sites need a second incision; cadaveric tissue or biodegradable scaffold alternatives; **secured to anterior rectus fascia on each side**; autologous fascia combines **compressive wrap with suspension.**
- **Excess tightness matters less** in neurogenic patients because most are managed by CIC.
- **Laparoscopic and robotic placement:** robotic operative time longer, but complications, stay and additional procedures similar; magnification aids deep pelvic access.

Results

- **Better and more extensive results in females;** some recent success in males; long-term success **varies widely.**
- **Circumferential wrap** may distribute compression over more bladder neck and posterior urethra, ± suspension; variation is too great to credit any modification.
- **Concomitant enterocystoplasty was the only predictor of success** — not age, sex, diagnosis, preoperative urodynamics or technique.
- **Erosion is rare with fascia** (unlike Silastic); revision uncommon; **does not preclude later AUS.**
- **Reasonable indication:** marginally competent bladder neck with deficient external sphincter in a child already undergoing augmentation and already on CIC.

Bladder Neck Injection

- **Minimally invasive** option for select sphincteric and bladder neck deficiency.
- **Polytetrafluoroethylene** (Vorstman et al., 1985) abandoned — **regional and distant particle migration.**
- **Current agents:** **polydimethylsiloxane** — solid silicone particles in a hydrogel carrier; **large particle size limits migration**; **dextranomer/hyaluronic acid** — microspheres in hyaluronic acid, **favoured in contemporary reports.**

Technique

- **Retrograde, antegrade or combined** endoscopic approach; **subepithelial placement ideal** — displaces epithelium into the lumen; **prior bladder neck surgery and scarring make it difficult**; circumferential injection achieves coaptation.

Results

- **Durability and efficacy lower than formal repairs** — the trade-off for minimal invasiveness; worthwhile after prior bladder neck surgery or to avoid reconstructive morbidity.
- **Success highly variable** (technique, material, definition, pathology, prior repair); long-term success modest; **continence deteriorates through about 18 months, then stabilises**.
- **Antegrade versus retrograde**: no significant difference; few become dry; improvement trends higher with antegrade.
- **Inferior to fascial sling** in augmented intrinsic sphincter deficiency at long follow-up.
- **Injection to rescue a marginal formal repair: no evidence of lasting value or of benefit from repeat treatment**.
- **Limited role, reserved for a select group whose defining criteria remain unknown**.

Artificial Urinary Sphincter

- **Prompt continence while preserving spontaneous voiding**. Introduced by **Scott et al. (1974)**.
- **AS800**: seamless pressurised balloon reservoir, non-kink tubing, improved cuff for bladder neck and proximal urethral coaptation; **pump lock allows deactivation and activation without surgery**. Alternatives have been explored.

Technique

- **Cuff at the bladder neck in all females and prepubertal males**; also most effective in pubertal and adult neurogenic males. **Bulbar urethra is an alternative only in males with mature spongiosum**.
- **Children do not outgrow a bladder neck cuff** through puberty; routine replacement unnecessary.
- **Cuff around a bowel segment is feasible but more prone to erosion → interpose omentum**.
- **Plane as for a fascial sling**; cuff snug, not tight.
- **Sterility paramount**: preoperative antibiotics and **confirmed sterile urine** — which then permit opening the bladder to aid dissection and placement.
- **Leave deactivated with cuff deflated after placement → pseudocapsule forms → erosion falls; activation delayed about 6 weeks**. A noncycled AUS occasionally provides enough resistance on its own.

Results

- **Continence must be weighed against cost**: mechanical malfunction needing reoperation, and catastrophic infection or erosion.
- **Device refinements have dramatically reduced secondary procedures**; 10–15-year and longer data show high continence with most sphincters functioning; **age at implantation does not affect continence**.
- **Reference long-term series (Indiana)**: high overall continence over a decade; **malfunction markedly lower with AS800** than earlier models; **erosion similar across models**.
- **CIC still needed in about three-quarters of neurogenic children** — and is performed successfully through the cuff.
- **Infection without erosion → remove, replace later**. Minimise infection: sterile urine, meticulous skin preparation, bowel preparation, perioperative parenteral antibiotics, copious antibiotic irrigation.
- **Erosion virtually eliminated when the cuff is the primary bladder neck procedure**.
- **Upper tract change and hydronephrosis after AUS**, occasionally renal insufficiency: **bladder neck occlusion unmasks or induces detrusor hostility** (falling compliance, rising overactivity); preoperative urodynamics identify only some → **anticholinergics for hyperreflexia, augmentation for poor compliance**; close observation always.

- **Timing with augmentation:** early data showed far more infection with simultaneous than staged procedures; later series showed low infection with simultaneous placement; long-term data show **no difference simultaneous versus staged**; placement is feasible even in total continent bladder and bowel reconstruction.
- **Ultimate benefit: high continence with potential spontaneous voiding. When CIC and augmentation are needed anyway, native-tissue continence avoids the lifelong risks of infection, erosion and mechanical failure.**

Urethral Lengthening

- **Young (1919):** excision of anterior bladder neck plus narrowing of the posterior remnant — **failed because the tubularised segment was unsupported within the bladder**; Dees and Leadbetter maximised muscle tone and extended the tube through the trigone.
- **Tanagho (1981):** cephalad-based anterior detrusor tube, circular fibres acting as sphincter — **he cautioned against it in the neurogenic population.**
- **Tube breakdown and poor results led to the flap-valve concept: Kropp and Angwafo (1986)** — anterior detrusor tube kept in continuity with the urethra, tubularised and **implanted into a submucosal trigonal tunnel**; effective, but **difficult catheterisation is a common, significant problem.**

Technique

Foley catheter, bladder filled to capacity → midline or low transverse incision → identify bladder neck by gentle catheter traction → isolate a **6 × 2 cm rectangular anterior flap** based on bladder neck and urethra, stay sutures, mobilise in continuity with proximal urethra → divide detrusor at the bladder neck (separating bladder from urethra), or leave muscle intact at **5 and 7 o'clock** → expose anterior vaginal wall (female) or seminal vesicles (male) → **tubularise posteriorly** around the catheter with continuous absorbable suture, **distal end interrupted** so excess can be trimmed → create a **capacious submucosal trigonal tunnel** between the orifices (Figure 48.3) → pull the tube through **straight, without curve or deviation** → **lateral anchoring sutures** at the bladder neck eliminate dead space → secure to trigone floor with interrupted absorbable sutures → close lateral wings incorporating the tube's adventitia for 2–3 cm anteriorly (watertight), often to the augmentation → the neourethra must reach the **true bladder lumen**, where pressure makes it a flap valve.

- **Not all ureters need cephalad reimplantation;** in a single-institution comparison, reimplantation was needed less often with Kropp than with Salle.
- **Belman and Kaplan (1989) simplification:** lateral and posterior muscle not incised, bladder and urethra not separated; flap tubularised over an **8 French (Fr)** catheter and laid in an **incised epithelial trough** rather than a tunnel; edges secured to the tube; suture line posterior against trigone; remainder covered by augmentation.
- **Whatever the modification, urethral catheterisation may remain problematic → prepare for a catheterisable abdominal channel.**
- **Vesicoscopic lengthening:** U-incision lengthening onto the dome, sparing the orifices and usable after failed repairs — **no long-term benefit.**

Results

- **High continence but difficult catheterisation, particularly in males** — most catheterised via an abdominal stoma. **New VUR** from lateral ureteric retraction → **leave the posterior bladder wall open and flat when receiving the augmentation.** Kropp reported fewer catheterisation problems and little new reflux.

- **Some never leak per urethra → no pop-off → upper tract deterioration or rupture if not catheterising reliably.** Snodgrass's modification deliberately gave a **shorter intravesical tunnel, allowing leakage with overfilling.**

Pippi Salle Procedure

- **Aim:** keep the benefits of Kropp while reducing catheterisation problems — an **anterior bladder onlay flap**; the **posterior neourethral wall is intact**, theoretically less catheter hang-up. Later modifications improved flap viability, reduced fistula and extended indications beyond neurogenic bladder.

Technique

Anterior full-thickness bladder wall flap **about 1 cm wide × 4–5 cm long** mobilised to the bladder neck, **0.1 cm of epithelial edges excised** to avoid overlapping suture lines → two parallel mucosal incisions down to muscle on the trigone from the bladder neck, creating a midline strip → flap sutured to strip (absorbable) to form the lengthened urethra → muscle beside the strip incised superficially and **approximated to flap muscle as a second layer** to prevent fistula (Figure 48.4) → lateral trigonal mucosa mobilised and closed over the neourethra → distally, bladder neck muscle **wrapped tightly around the urethra**, incorporating part of the flap for a watertight seal → proximally, neourethra extends **well into the lumen** → remaining bladder left open if augmenting.

Results

- **Early complications:** persistent incontinence, **urethrovesical fistula at the flap base**, partial necrosis of the intravesical neourethra → **widen the flap base and trim epithelial edges generously.**
- **Secondary Salle** after prior bladder surgery: **lateralise the flap** away from the old midline suture line.
- **Catheterisation difficulty less than Kropp and rarely persists; continence lower than Kropp in most series**, with night-time dryness notably poorer.
- **Miniature lengthening:** shorter (**3 cm**) and narrower (**8 Fr**) neourethra — raises resistance, uses less bladder, avoids reimplantation; good continence at medium follow-up.
- **Kropp versus Salle:** similar dry intervals without further bladder neck surgery; **high rate of delayed augmentation if not augmented at the bladder neck procedure**; high rate of secondary surgery unrelated to the bladder neck in both.

KROPP VERSUS PIPPI SALLE

	Kropp	Pippi Salle
Construct	Tubularised anterior flap, tunnelled under trigone	Anterior flap onlaid on trigonal strip
Posterior wall	Tube suture line	Intact native strip
Flap	6 × 2 cm	1 × 4–5 cm
Continence	Higher	Lower, especially at night
Catheterisation	Difficult (males)	Easier
Signature complication	Catheter hang-up; new VUR	Fistula at flap base ; tip necrosis
Reimplantation	Less often	More often

- **Pattern:** onlay trades some continence for catheterisability. **Trap:** either operation abolishes voiding and pop-off; both carry high delayed augmentation if the bladder is not augmented concurrently.

Bladder Neck Division

- **The ultimate outlet procedure:** bladder neck separated from the urethra; **must be accompanied by a continent abdominal stoma and performed only in reliable catheterisers.**

- **No urethral leakage or pop-off** → **increased risk of upper tract deterioration or rupture** if emptying lapses.
- **Seldom primary**; considered when other repairs fail.
- **Moves reconstruction into continent urinary diversion**: the bladder neck is physiologically abandoned; the bladder may form part of the reservoir.

Aggressive mobilisation of bladder from urethra → posterior lip of bladder neck and trigone **rolled anteriorly** and widely separated from the distal urethra → **multilayer distal closure** → **interpose omentum or rectus abdominis muscle**. Omitting these steps → **fistula and urethral leakage**; with aggressive mobilisation the fistula rate is low. Despite loss of secondary access, **long-term success is achieved in almost all**.

KEY POINTS: BLADDER NECK RECONSTRUCTION

- Any repair other than AUS, particularly combined with cystoplasty, is likely to make spontaneous voiding inadequate.
- Young-Dees-Leadbetter repair in spina bifida has variable success related to the underlying neurogenic dysfunction.
- Fascial slings have been used more extensively and with better results in females than in males, although results vary widely.
- Bladder neck bulking agents have not yet shown a lasting effect on severe neurogenic sphincter incompetence.
- A functioning, well-positioned AUS provides good resistance and is best reserved for patients who empty adequately by voiding.
- Urethral lengthening uses a flap valve that disrupts any chance of voiding and may make urethral catheterisation difficult.
- Most bladder neck repairs in the neurogenic population succeed better when combined with augmentation.
- Bladder neck repair may unmask or create new bladder hostility; careful follow-up is mandatory.

Concept — outlet resistance and detrusor safety are inversely coupled: every increment of continence gained at the bladder neck is paid for in lost voiding, lost pop-off and risk of an unmasked hostile bladder, which is why neurogenic outlet repairs succeed mainly with augmentation and all demand years of surveillance.

BLADDER RECONSTRUCTION

Cutaneous Vesicostomy

- **"Vents" the bladder**, sparing the upper tracts from high intravesical pressure.
- **Indications**: worsening upper tract dilatation, recurrent febrile UTI with high-grade reflux, **high-pressure bladder despite maximal medical therapy**; temporary or permanent.
- **Typically in young patients**; a safe bridge to definitive reconstruction.
- **Disadvantages**: stenosis, prolapse, incontinence — harder to manage as the child grows.

AUGMENTATION CYSTOPLASTY

- **Initial approach is the same regardless of segment.**

- **Preoperative cystoscopy** for unsuspected anatomic abnormality; **leave the bladder full** if other bladder procedures (reimplantation) follow, otherwise empty it for peritoneal access.
- **Midline incision generally preferred**; lower transverse if no prior abdominal surgery; laparoscopic bowel mobilisation allows a smaller, lower incision.
- **Associated bladder procedures before opening the peritoneum** to minimise third-space loss.
- **Totally laparoscopic ± robotic: 12-mm camera port** peri- or supraumbilical; up to **four 5–8-mm working ports**, one larger for an endostapler; assistant ports optional; fourth robotic arm aids retraction and suture passage.

Management of the Native Bladder

- **Formerly: supratrigonal excision** of the "diseased" bladder leaving a small cuff (Figure 48.6). **Now: preserve the native bladder but open it widely — a narrow-mouthed anastomosis makes the segment behave as a diverticulum** (Figure 48.5).
- **Sagittal bivalving incision**: from several centimetres cephalad to the bladder neck anteriorly to just above the trigone posteriorly — easier anastomosis, native bladder adds capacity; a **stellate** opening (second, transverse incision into both halves) gives greater circumference.
- **Penile or perineal pain in sensate males** after preservation has occasionally required secondary bladder excision — **not frequent enough to justify routine excision**.

Management of Intestinal Segments

- **Detubularise on the antimesenteric border and reconfigure into a spherical shape**: maximal volume per surface area, **blunted bowel contractions**, improved capacity and compliance.
- **Intact tubular segments generate contractions of 60–100 cm H₂O**; detubularisation markedly lowers peak contractile pressure in both right colon and ileum, **more so in ileum**; a **shorter segment achieves the same capacity**.
- **Detubularisation and reconfiguration should always be performed**.
- **Segment length**: mathematical models exist but are cumbersome; **20–40 cm of ileum or about 20 cm of colon** typically; **less bowel for a moderate-volume bladder with high pressure from uninhibited contractions than for a tiny one**.
- **Err on the side of too large rather than too small**; allow for urine output — damaged upper tracts may produce huge volumes.

Ileocystoplasty

- **Goodwin et al. (1959)**: among the first to show the many ways of anastomosing a detubularised, maximally spherical ileal patch to the bladder.

Technique

Select ileum **at least 15–20 cm proximal to the ileocaecal valve** (terminal ileum is physiologically unique) → **20–40 cm** segment per patient size, native capacity and target capacity; an **extra isoperistaltic tail** for short ureters requires an **ileal nipple valve** against reflux (Kock or hemi-Kock) and up to **60 cm** of small bowel → confirm an **adequate mesentery reaching the bladder without tension** → clear mesentery at each end to create windows, divide bowel → hand-sewn or stapled ileoileostomy → **close the mesenteric window** (internal hernia) → **irrigate the segment with 0.25% neomycin** → open on the antimesenteric border (Figure 48.7A) → fold in a **U** (commonest), or **S** or **W** for longer segments, or as a cup patch → running absorbable full-thickness ileum-to-ileum suture inverting mucosa → anastomose to bladder **beginning posteriorly**, one or two layers, **absorbable suture, mucosa inverted** → **suprapubic tube through the native bladder** →

complete anterior anastomosis → **pelvic drain via a separate stab incision**, removed promptly if not draining urine, **especially with a VP shunt**.

- **Permanent sutures are never used in any cystoplasty — they are a nidus for stones.**

Cecocystoplasty and Ileocecocystoplasty

- **Couvelaire (1950)** described caecal augmentation; **caecocystoplasty is now uncommon, replaced by ileocaecocystoplasty**: caecum opened and reconfigured for augmentation, leaving ileum to reach the ureters or form a continent stoma; or ileum opened as a patch on the caecum.

Technique

Mobilise caecum and right colon along the **white line of Toldt to the hepatic flexure** → take **15–30 cm of terminal ileum** per technique → confirm tension-free reach before division (Figure 48.8A) → irrigate with neomycin → open on the antimesenteric border **through the ileocaecal valve for its entire length** → standard: **equal ileal and caecal lengths**, borders sewn and folded into a **cup** (one or two layers, absorbable), opening left wide for the bivalved bladder → more volume: longer ileum folded before joining caecum; **Mainz ileocaecocystoplasty uses ileum twice the caecal length** (Figure 48.9, Thüroff, Alken and Hohenfellner, 1987; ureters can be reimplanted into the opened caecum) → close mesenteric window; suprapubic tube through native bladder and abdominal wall.

Appendix

- **Advantage of ileocaecocystoplasty: the appendix** for a reliable continent stoma — removed with a small caecal cuff and tunnelled into native bladder or a caecal taenia, or **left in situ with its base tunnelled beneath a taenia**; otherwise appendectomy.

Ileocecal Valve

- **Used extensively in adults, less in children**, because most augmented children have **neurogenic bowel as well as bladder: removing the valve can cause intractable diarrhoea** → avoid unless other advantages outweigh diarrhoea and faecal incontinence risk.
- **Advantages**: easy **antireflux tunnels into the caecal taenia**; an **ileal tail bridges a short ureter with the imbricated valve as antireflux**; the same imbrication creates a **continent stoma as in the Indiana pouch**.

Sigmoid Cystoplasty

- **Lemoine (1912)** first reported; still used.
- **Strong unit contractions** → **detubularisation and reconfiguration are imperative** for compliance.

Technique

Identify and mobilise **15–20 cm of sigmoid** → **transilluminate the mesentery** to identify the arcade → confirm tension-free reach → divide between clamps or staples; **colocolostomy** (Figure 48.10A) → pack the abdomen against contamination from the open segment → detubularise and reconfigure per preference → anastomose to the bivalved bladder as for ileum → **large suprapubic tube, or intravesical 10-mm flat Jackson-Pratt drain**, brought through native bladder and secured at bladder and skin.

Reconfiguration of sigmoid

- **Mitchell (1986)**: **close both ends, open longitudinally opposite the blood supply**; fits the bivalved bladder sagittally or coronally, may be rotated 180°.
- **U-fold as for ileum**: **more radical reconfiguration, better disruption of unit contractions**, may lower pressure; needs a slightly longer segment.

Postoperative Management

Early Management

- **Similar regardless of segment.**
- **Nasogastric decompression historically routine; may be unnecessary after ileocystoplasty** → surgeon preference.
- **Fluid and electrolytes:** third-space losses may be significant.
- **Continuous suprapubic drainage;** mucus may occlude it → **irrigate at least three times daily** and whenever drainage slows.
- **Extravesical drains** removed after several days if no urine; **earlier with a VP shunt.**
- **Static cystogram at about 3 weeks;** no leak → clamp the suprapubic tube, remove after a few days of successful channel catheterisation.
- **CIC every 2–3 hours by day and once or twice at night** initially → interval lengthened over weeks, **never beyond 4–5 hours by day.**
- **Patients without neurologic impairment** may later attempt voiding; all check residuals and continue CIC if significant.

Late Management

- **Protocols are not well defined.**
- **Renal ultrasound at 3 months, 6 months and 1 year;** some centres perform **urodynamics within 6 months.**
- **Electrolytes, BUN, creatinine and urine culture at 3 and 6 months,** then yearly or when symptomatic; ultrasound and serum chemistry yearly thereafter.
- **Not every positive culture on CIC needs treatment; treat symptomatic cystitis and urea-splitting organisms.**
- **Cystoscopic surveillance:** no consensus; possibly higher yield in higher-risk groups — **valve patients immunosuppressed after transplantation, exstrophy (inherent adenocarcinoma risk), gastrocystoplasty; not shown to be cost-effective;** the Mayo Clinic group considers it unnecessary.

Results and Complications of Augmentation Cystoplasty

- **Judge augmentation two ways:** (1) **cost to the gastrointestinal tract** of removing a segment — more than rare gastrointestinal problems would be prohibitive, however good the urinary result; (2) **effect on the bladder** — the primary goal is a **compliant reservoir**, so the **main outcome measures are storage pressure and capacity;** everything else is a complication arising because **bowel is not a perfect physiologic substitute for bladder.**
- **Further surgery** is needed in **up to one-third;** a modern cohort puts the **10-year risk of additional surgery near one-half.**

Gastrointestinal Effects

- **Bowel obstruction is uncommon,** at a rate **equivalent to conduit or continent diversion;** prevented by delicate handling, closure of mesenteric windows and elimination of internal hernia sites; segment differences inconsistent, probably not significant.
- **Chronic diarrhoea after augmentation alone is rare;** the ileal lengths used are rarely problematic; colon rarely changes bowel function.
- **Removing the ileocaecal valve is most likely to cause diarrhoea:** some neurogenic children **depend on constipation for faecal continence;** loss of the valve shortens transit time and allows **bacterial backflow**

into ileum, disturbing fat and vitamin B₁₂ metabolism; chronic diarrhoea follows in a notable minority of neurogenic patients, fewer when carefully selected.

- **Ileum is the sole site of vitamin B₁₂ absorption** → distal ileal loss → **B₁₂ deficiency and megaloblastic anaemia**; the **terminal 15–20 cm is never used**, although problems can arise even when it is preserved.
- **Risk scales with ileal length**: deficiency is common after a Kock pouch; augmentation uses **less than half the Kock length**, so deficiency is unlikely — but **body stores last years**, demanding long observation; **low or low-normal B₁₂ is frequent years after ileocystoplasty and responds to oral replacement**.

Bladder Compliance After Augmentation

- **Detubularisation and reconfiguration** are the defining early lesson.
- Some experienced surgeons consider **ileum superior for compliance** (uncontrolled); one report favoured a longer U-shaped colon; **using a segment well matters more than which segment is chosen**.
- **Pressure problems arise from uninhibited contractions, apparently in the bowel segment; failure to reach adequate capacity or a flat tonus limb is extremely rare without technical error**.
- **Pattern: rhythmic, sinusoidal contractions**, sometimes of increasing amplitude (Figure 48.11).
- **Low-amplitude contractions arising late, near capacity**, may be insignificant; **early high-pressure contractions → persistent incontinence, delayed perforation, hydronephrosis, VUR → repeat urodynamics; compliance is never assumed after augmentation**.
- **Ileum least likely and stomach most likely** to show remarkable urodynamic abnormality; colon (caecum or sigmoid) still generates more pressure than ileum despite detubularisation, although colonic contractions may decrease with time; ileal reservoirs show lower basal pressure and less motor activity; **significant contractions — above 40 cm H₂O at volumes below 200 mL — were not seen after ileal augmentation** but persisted after some colocystoplasties.
- **No study controlled segment size or technique**; a canine model with **identically sized segments showed no difference** between segments.
- **Gastrocystoplasty: rhythmic contractions are frequent**; antrum or a larger gastric segment may lessen them.
- **Clinically significant contractions affect a small minority; reaugmentation** is required in a small minority long term — **true failures** of capacity and compliance. At Indiana, **sigmoid** (closed at the ends and not otherwise reconfigured) > stomach > ileum; other series found stomach more likely than colon to need reintervention.
- **Figure 48.11: sinusoidal contractions after gastric augmentation**; after secondary ileal augmentation, contractions persist but at **lower pressure and later in filling**.

Metabolic Complications

Chloride absorption and acidosis

- **First recognised metabolic complication: hyperchloraemic metabolic acidosis after ureterosigmoidostomy** (Ferris and Odel, 1950) — fatigue, weakness, anorexia, polydipsia.
- **Mechanism** (Koch and McDougal, 1985): **resorption as ammonium → chronic acid load**.
- **With normal renal function: a rise in chloride and fall in bicarbonate is near universal, but frank acidosis is rare**; nonetheless, treatment-requiring acidosis occurs despite normal renal function and is debilitating if unrecognised.
- **Increased urinary acid load with wasting of bony buffers, even without frank acidosis → bone demineralisation, possibly retarded growth**.
- **Treat acidosis with bicarbonate**; **arterial blood gas base deficit is more sensitive** than serum bicarbonate or chloride and allows early treatment before demineralisation.

- **Severe acidosis: block chloride transport with chlorpromazine and nicotinic acid.**

Jejunum

- **Rarely used;** stores urine with a unique **hyponatraemic, hypochloraemic, hyperkalaemic metabolic acidosis, often with significant hypovolaemia.**

Stomach and alkalosis

- **Gastrocystoplasty avoids hyperchloraemic acidosis,** even with impaired renal function — but its **secretory mucosa** causes two unique complications.
- **Hypokalaemic, hypochloraemic metabolic alkalosis,** typically **during acute gastrointestinal illness with dehydration** — ongoing gastric chloride loss with reduced intake; often needs admission, may recur; **patients with renal insufficiency — the main indication for stomach — are at greatest risk.**
- **Inappropriate gastric secretion is the primary mediator:** persistently raised fractional excretion of chloride despite profound hypochloraemia; treated with **H₂ blockers or H⁺/K⁺ pump inhibitors**, rarely partial resection of the gastric segment. All families are warned; it occurs intermittently in a minority.
- **Haematuria-dysuria syndrome:** acid urine irritates native bladder and exposed skin, worst with **intact sensation** (Mitchell's group).
- **Composite stomach with ileum or colon** → a **more metabolically neutral reservoir;** used in the most complex patients.

METABOLIC CONSEQUENCES BY SEGMENT

Segment	Derangement	Setting and management
Ileum, colon	Hyperchloraemic metabolic acidosis (ammonium, chloride absorption)	Worse with renal impairment; bicarbonate; base deficit to detect; chlorpromazine/nicotinic acid if severe
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis + hypovolaemia	Reason jejunum is avoided
Stomach	Hypokalaemic, hypochloraemic alkalosis; haematuria-dysuria	Acute gastrointestinal illness, renal insufficiency; H ₂ blocker or proton pump inhibitor
Composite stomach + bowel	Near neutral	Most complex patients

- **Pattern:** absorptive segments load acid; secretory stomach loses acid. **Trap:** stomach avoids acidosis in renal failure, yet those same patients are the ones most at risk of its alkalosis.

Patient growth and bone

- **Delayed linear growth** reported after cystoplasty without gross biochemical abnormality, in uncontrolled series; growth in myelodysplasia is hard to predict.
- **Matched exstrophy series: decline in height percentile after augmentation** — worrisome; mechanism likely **subclinical acidosis**; one series found **no adverse effect** in exstrophy.
- **Bone mineral density:** reduced in some series, not others; augmentation effect must be separated from the underlying pathology; better metabolic analysis may permit early treatment.

Mucus

- **Mucus production persists** → impedes drainage (small-calibre paediatric catheters), **nidus for infection and stones; increases with cystitis.**
- **Colon > ileum > stomach** (canine work, confirmed clinically — gastric augmentations rarely need irrigation).

- Ileal villous atrophy is documented, but **falling production over time is unproven**; colonic decline inconsistent; no colonic glandular atrophy histologically.
- **Routine daily irrigation minimises UTI and calculi.**

Urinary Tract Infection

- **Bacteriuria is common**, particularly with CIC; spontaneously voiding neobladder patients often keep sterile urine → **CIC is a prominent factor**; persists despite daily oral antibiotics or antibiotic irrigation.
- **Segment trend**: more bacteriuria with sigmoid than ileum; CIC-dependent patients more regardless of segment; recurrent symptomatic cystitis **ileum > sigmoid > caecum > stomach**, febrile UTI the same trend without significance.
- **Warn all families to expect some cystitis.**
- **Pyelonephritis, once upper tract problems are corrected, is similar to conduit diversion, refluxing or not.**
- **Immunocompromise**: occasionally more problematic, not consistently.
- **Treat bacteriuria**: significant symptoms (incontinence, suprapubic pain) and **any urea-splitting organism**; **may treat**: haematuria, foul urine, markedly increased mucus.
- **Prevention**: regular CIC (avoiding high reservoir pressure) and complete emptying — harder through a **nondependent abdominal stoma** but achievable with effort; **clean, not sterile, technique.**

Calculi

- **Common long-term complication**; rates vary widely between series for unclear reasons; **catheterisation through an abdominal wall stoma carries the highest risk** (incomplete emptying).
- **Mostly struvite; bacteriuria is the key risk factor** → treat urea-splitting infection.
- **Empty completely at every catheterisation**; stones in a voider → **re-evaluate emptying**; the association with stasis is well established.
- **Daily irrigation of at least 240 mL** markedly reduced recurrence in prior stone formers and, applied routinely, reduced overall stone risk (Husmann, 2018); Indiana and others advise irrigation several times daily.
- **No difference between small and large bowel**; **struvite less likely after gastrocystoplasty** (less mucus, acid limits bacteriuria); **uric acid calculi** occur after gastrocystoplasty.
- **Any foreign body is a nidus — no permanent sutures or staples.**
- **Bladder stones and intestinal interposition may be independent risk factors for upper tract stones** in neuropathic bladder.
- **Low urinary citrate** in patients with and without stones; poor emptying and mucus judged more important.

Tumor Formation

- **Ureterosigmoidostomy: adenocarcinoma at the ureterocolonic anastomosis; mean latency about 26 years (range 3–53)** (Husmann and Spence, 1990); risk estimated at **about 7000-fold** that of age-matched controls (Eraklis and Folkman, 1978); a Swedish undiversion cohort showed markedly increased invasive colorectal cancer with deaths.
- **Proposed mechanisms: N-nitroso compounds** from mixed urine and faeces (also found in conduit and augmentation urine; probably enhancers rather than sole cause); **anastomotic inflammation → growth factors → proliferation.**
- **Adenocarcinoma in augmented bladders** after ileal and colonic cystoplasty (Filmer and Spencer, 1990); **relatively high incidence after gastrocystoplasty**, including fatal cases within the gastric segment; rat models: hyperplasia with every segment, none of higher risk.

- **Earliest tumour 4 years after augmentation**; given long latency, short follow-up cannot assess risk; a multi-institutional series found tumours (mostly adenocarcinoma) **decades after** conduits, gastrocystoplasty, augmentation and Indiana pouch.
- **Counsel: risk highest after ureterosigmoidostomy and gastrocystoplasty**; the increase may reflect an **inherent cancer risk of the underlying congenital abnormality** rather than augmentation. A spina bifida mortality analysis found no deaths from malignancy or perforation at medium follow-up.
- **Yearly endoscopic surveillance may eventually be performed**; when it should start is undefined.
- **Transitional cell carcinoma, hyperplasia and dysplasia near the anastomosis**; **transitional cell carcinoma after augmentation is aggressive, often metastatic at presentation**; perianastomotic urothelium is genetically unstable.
- **Nephrogenic adenoma** — benign, probably **the commonest tumour after cystoplasty**, and frequently recurrent.

Delayed Spontaneous Bladder Perforation

- **Presentation**: ill with abdominal pain, distension, fever; **sepsis common**; nausea, falling urine output, shoulder pain from **diaphragmatic irritation**; occasionally an asymptomatic pelvic mass.
- **Neurogenic patients have impaired lower abdominal sensation → present late → severe sepsis and death**; gastrocystoplasty patients **present promptly** because of acid irritation.
- **High index of suspicion. Contrast cystography is diagnostic in most** — thorough technique, full filling and a **postdrain view** (Figure 48.12); significant **false negatives** → **ultrasonography or computed tomography (CT)** if the cystogram is **negative** in a suspected child.

Etiology

- **Cause unknown. Traumatic catheterisation** proposed (perforation occurs in non-augmented bladders on CIC), but sites vary even within one patient and perforation occurs in non-catheterisers → **unlikely sole cause**. Also proposed: **adhesion-related shearing** with filling and emptying; **transmural infection**.
- **Histology: necrosis, vascular congestion, haemorrhage, haemosiderin → chronic bowel wall ischaemia**; **chronic overdistension** causes it — high intravesical pressure reduces perfusion experimentally, most at the **antimesenteric border**.
- **Uninhibited contractions** are associated; **high outflow resistance maintains pressure instead of venting it by leakage**, increasing ischaemia; hyperreflexia alone is not the cause (perforation was unrecognised before detubularisation, when contractions were commoner), but **reconfigured bowel may be more ischaemia-prone if high pressure persists**; **missed CIC** exacerbates.
- **Extraperitoneal approach does not lower risk** versus intraperitoneal.
- **Most perforations occur in myelodysplasia**; lower in other diagnoses; role of neurogenic dysfunction unclear.
- **Field effect on the whole segment → recurrence is significant** → after repeated perforation, **replace the segment**.

Incidence

- **Early anastomotic leaks are rare** and reflect technical error or poor healing; **delayed perforation occurs within the bowel segment** and reflects long-term urine storage in bowel.
- Indiana found **sigmoid** the only significant risk factor, years after augmentation; other large sigmoid series found little perforation, and another found more with ileum → **no segment is consistently at higher risk**; **multiple factors**.
- **Bladder outlet procedures double perforation risk**.

- **Counsel an expected risk of about 9%** (Indiana); in a transitional population incidence was lower, with small mortality, and **substance abuse and catheterisation noncompliance were directly associated** (Husmann, 2017).

Treatment

- **Standard: immediate surgical repair.**
- **Conservative management** (catheter drainage, antibiotics, serial examination) reported successful, but few cases were radiographically proven; one series supports it in the **absence of clinical deterioration, severe peritonitis and extensive extravasation** — maximal bladder drainage plus **image-guided percutaneous drains** for loculated collections; **many still need eventual surgery.**
- **Only in a stable patient with sterile urine, with a very low threshold for surgery** — delay leads to sepsis and death.

Pregnancy

- **Three questions:** effect of the enlarging uterus on the **pedicle**; management of **bacteriuria**; **need for caesarean section.**
- **Pedicle:** usually deflected laterally, occasionally draped over the uterus; the uterus still rises normally; **no reported loss of an augmentation from pedicle stretch.**
- **Indiana experience: high complication rate** — new or worse hydronephrosis very common, often needing **percutaneous nephrostomy, preferred over stent** (altered anatomy, mucus plugging); difficult catheterisation, incontinence, UTI; **urology present at every caesarean**, with frequent intraoperative complications, commonest **cystotomy during emergency caesarean**; urinary fistula.
- **UTI risk factors:** ureteral dilatation, increased residual, diminished upper tract tone; asymptomatic bacteriuria is normally treated in pregnancy, but **no consensus for CIC after cystoplasty**; antibiotics must be **non-teratogenic.**
- **Vaginal delivery is possible** after augmentation and other reconstruction; continent catheterisable ileocaecal diversion: no major complications; **late catheterisation difficulty** can occur with any continent abdominal stoma.
- **Caesarean is not universal**; authors' bias: **after extensive bladder neck repair, consider caesarean**, particularly if labour progresses slowly; with an **AUS, a long difficult labour raises erosion concern.**
- **At caesarean: protect the augmentation or continent stoma and its pedicle**; collaterals usually sustain them if injured; **postpartum urodynamics** for augmented patients; the **reconstructive urologist familiar with the anatomy should be present.**

Choice of Segment and Approach

- **No single segment is perfect for all**; all have been used with good results; unremitting problems are rare with appropriate selection; **no segment has a clear overall advantage.**
- **Diagnosis, anatomy and physiology may favour one**; surgeon familiarity and confidence matter; **choose the segment to fit the patient, then use it correctly.**
- **Ileum is preferred** when there is no clear reason to use another.
- **Stomach is reserved** for very select children: **renal insufficiency with acidosis, short gut syndrome, heavy irradiation** — weighing its complications.
- **Sigmoid** is used in select patients without reservation.
- **Minimally invasive (usually robotic) augmentation** evolved from autoaugmentation to full intracorporeal enterocystoplasty; operative times vary with experience and patient factors; **early outcomes mirror open surgery**; faster recovery and better cosmesis; cost-effectiveness in the neurogenic population unproven.

BOWEL SEGMENTS FOR AUGMENTATION COMPARED

Segment	Length	Strengths	Liabilities
Ileum	20–40 cm, ≥15–20 cm from ileocaecal valve	Default ; least urodynamic abnormality; lower basal pressure	Acidosis; B ₁₂ if distal; hardest antireflux
Ileocaecal	15–30 cm ileum + caecum (Mainz: ileum 2×)	Taenial reimplant; appendix; valve imbrication	Diarrhoea in neurogenic bowel ; colonic mucus
Sigmoid	15–20 cm	Taenial reimplant; no small-bowel loss	Strong unit contractions ; most mucus; reaugmentation (Indiana)
Stomach	—	No acidosis ; least mucus; fewer struvite stones	Alkalosis ; haematuria–dysuria; rhythmic contractions; tumour
Jejunum	—	—	Hyponatraemic hyperkalaemic acidosis ; rarely used

- **Pattern**: each segment's liability follows from its physiology — absorption, secretion, contractility or its role in bowel continence. **Trap**: expecting a segment to be chosen on outcome superiority; technique matters more than segment.

A Decreasing Necessity?

- **Goal of every paediatric urologist: minimise the number needing cystoplasty** — newer medical regimens, **botulinum toxin A injection**, **neuromodulation**.
- **Earlier, more aggressive treatment** limits bladder insult and maximises recovery; **early urodynamics in posterior urethral valves** identifies treatable bladder dysfunction.
- **Kaefer et al. (1999a)** — the most compelling evidence: **hostile neurogenic bladders treated immediately at diagnosis needed augmentation in 17% versus 41% when treated expectantly**; no urodynamic data, possible lag-time bias.
- **No randomised trials** → the benefit of early treatment remains unproven; UMPIRE may clarify. Early bladder management initially showed no fall in augmentation; re-evaluation suggests a **fall of about a quarter over a decade**. Improvements will reduce, not abolish, cystoplasty.

Understanding the Patient-Centered Impact of Incontinence and Reconstruction

- **Goals: protect renal function and improve continence** — improved health-related quality of life is assumed, not demonstrated.
- **Early small series showed no health-related benefit on questionnaires despite good surgical outcomes**; social impact of a channel versus the urethra is little studied; the literature is surgeon-centred.
- **Emerging data**: moderate perceived benefit after reconstruction; **faecal incontinence affects quality of life more uniformly across childhood and earlier than urinary incontinence**; urinary incontinence limits social participation; incontinence and its management are distressing and unpredictable.
- **Next step: condition-specific tools**, such as web-based goal attainment for catheterisation compliance and continence tracking in spina bifida.

Ileovesicostomy

- **Incontinent alternative**; counselling of parents and older patients is essential.
- **Candidates**: patients **mentally, physically or socially unable to catheterise a continent channel**; also **hyperreflexic detrusor with small capacity**.
- **Versus vesicostomy**: dry per urethra, **bag appliance** instead of constant leakage and diapers. **Versus continent diversion: low-pressure reservoir → greatly reduced perforation risk**, may avoid sphincteric surgery, no catheterisation, **convertible to continent diversion**.

- **Shared risks:** bowel complications, stomal stenosis or prolapse, stones, UTI, upper tract deterioration.

Technique and Results

- **10–15 cm of ileum**, as for a conduit, anastomosed **isoperistaltically to the bladder dome** and brought out as a **rosebud stoma** — a passive, path-of-least-resistance channel.
- **Paediatric data scant:** complication profile like adults; long-term effects unassessed; **complications rise with follow-up length and body mass index (BMI)**; obesity is common in spina bifida at transition and associated with reoperation → **regular follow-up** to protect the upper tracts.

VESICOSTOMY, ILEOVESICOSTOMY AND CONTINENT RECONSTRUCTION

	Vesicostomy	Ileovesicostomy	Continent reconstruction
Continence	None (diapers)	Dry per urethra; bag	Continent
Catheterisation	No	No	Obligatory
Reservoir pressure	Low	Low	Depends on emptying
Perforation risk	—	Low	Present
Bowel used	No	10–15 cm ileum	Usually
Typical patient	Infant, bridge	Cannot catheterise	Committed catheteriser

KEY POINT: ILEOVESICOSTOMY

- Ileovesicostomy is viable for a carefully selected group but carries increased long-term complication risk if not followed appropriately.

Alternatives to Gastrointestinal Cystoplasty

- **Motivation:** complications of bowel; efforts span synthetics, autologous grafts, **autoaugmentation** (bladder diverticulum) and neural stimulation — **none has stood the test of time against intestinal cystoplasty**.
- **Ideal augmenting tissue: transitional epithelium** (relatively impermeable, no metabolic change), **no mucus, no added tumour risk**.
- **Ureterocystoplasty:** transitional epithelium **with muscle backing**. **Autoaugmentation:** transitional mucosa eventually **backed by collagen**.

Ureterocystoplasty

(Figure 48.13) Nephrectomy → bladder bivalved, **posterior incision carried off the midline into the ureteral orifice; ureter not detached from the bladder** → ureter opened **opposite its main blood supply** → reconfigured as for intestinal cystoplasty → anastomosed to the bladder.

Autoaugmentation

(Figure 48.14; Cartwright and Snow, 1989) Detrusor incised → detrusor stripped and excised from mucosa → detrusor **anchored bilaterally (psoas hitch)** so that the **mucosa bulges** as a diverticulum with filling.

Seromuscular Enterocystoplasty

(Figure 48.15; Buson et al., 1994) Detrusor incision as in autoaugmentation → bulging bladder mucosa **covered with a demucosalised sigmoid segment** sutured to bladder muscle → space between bladder mucosa and sigmoid drained.

URETEROCYSTOPLASTY VERSUS AUTOAUGMENTATION

	Ureterocystoplasty	Autoaugmentation
Lining	Urothelium	Urothelium
Backing	Ureteric muscle	Collagen (eventually)
Requirement	Remarkably dilated ureter, nephrectomy	Native bladder mucosa
Limitation	Inadequate volume if ureter not very dilated	Seldom adds significant capacity; compliance may not last

Bladder Regeneration

- **Nearly two decades of effort** (tissue-engineered scaffolds, bioprinting); a bladder with specialised urothelium and contractile muscular stroma **has not been achieved**.
- **Atala et al. (2006)**: first seeded biodegradable construct for neurogenic augmentation → industry-supported multi-institution trial: **no clinically or statistically significant improvement in compliance or capacity**.
- **Remains experimental**, not for human use outside well-designed trials.

KEY POINTS: AUGMENTATION CYSTOPLASTY

- Any gastrointestinal segment should be reconfigured and widely anastomosed to the bladder to maximise capacity and compliance.
- Failure to achieve adequate compliance is rare and usually related to rhythmic pressure contractions.
- Gastrointestinal effects are rare, although chronic diarrhoea may follow use of the ileocaecal segment in the neurogenic population.
- Ammonium resorption can cause metabolic acidosis, particularly with renal insufficiency.
- Bacteriuria is common after cystoplasty and does not always require treatment.
- Bladder calculi are common after augmentation; the incidence can be reduced with daily irrigation.
- Aggressive adenocarcinomas have been reported as early as 4 years after augmentation, but not with the frequency seen after ureterosigmoidostomy.
- Delayed perforation of the bowel segment is to be expected in a minority; the reported excess in colonic segments has not been consistent across series.
- Good early care of the underlying bladder dysfunction may lower the need for augmentation.

KEY POINTS: ALTERNATIVES TO GASTROINTESTINAL CYSTOPLASTY

- Ureterocystoplasty avoids bowel complications but may not provide adequate volume unless the ureter is remarkably dilated.
- Autoaugmentation seldom adds significant capacity and may not improve compliance long term.
- Bladder regeneration remains experimental and not ready for human use outside well-designed trials.

Concept — augmentation buys compliance by **importing an absorptive, secretory, contractile organ into the urinary tract**: every late complication — acidosis, alkalosis, mucus, stones, infection, tumour, perforation — follows from bowel not being bladder, and every mitigation (reconfiguration, irrigation, complete emptying, surveillance) targets one of those properties.

CONTINENT URINARY DIVERSION

- **Frequency depends on definition.** Cystectomy for tumour is far rarer than in adults; after cystectomy for cancer or severe congenital bladder anomaly, a **pure continent diversion (Kock or Indiana pouch)** gives results equivalent to adults.
- **Orthotopic neobladder is rare in children:** neurogenic or anatomic outlet problems prevent spontaneous voiding; an occasional neurogenic child is a candidate, but adequate voiding is uncertain.
- **Broader definition:** augmentation + continent abdominal stoma + an outlet procedure (bladder neck division typically counts) — in complex patients, often after many previous operations.

Considerations

- **Total replacement needs far more bowel** than augmentation: **about 40 cm of ileum for a Kock reservoir versus about 20 cm for augmentation**; the **entire right colon with hepatic flexure for an Indiana pouch versus 15–20 cm for colcystoplasty** → the **native bladder is used if it provides any significant volume**, even if the outlet must then be repaired.
- **Imbricated ileocaecal valve and terminal ileum:** simple, reliable efferent limb in adults and children; yet, despite contrary reports in select patients, **concern about faecal incontinence persists in neurogenic patients** → **flap valves (appendix, tapered bowel)** are increasingly used.
- **Native urethra is ideal for CIC but not always feasible:** catheterisation difficulty after bladder neck reconstruction; **neurologic limitation of access, especially the wheelchair-bound child**; in neurologically intact children, **urethral sensation causes discomfort** that undermines the schedule → a **continent catheterisable stoma is an adequate and sometimes more reliable alternative**.

Continence Mechanisms and Catheterizable Stoma

Ureterosigmoidostomy and Its Variants

- **Advantage:** urine evacuated with stool. **Complications:** hyperchloraemic acidosis, infection, hydronephrosis, colonic malignancy → disfavour, particularly in the United States.
- **Rarely performed as originally described;** acidosis and upper tract morbidity rise with time; the **adenocarcinoma risk commits a child with a long life expectancy to lifelong surveillance for an avoidable tumour** → unlikely to regain acceptance in children.
- **Sigma rectum (Mainz II pouch)** — the most basic modification: antimesenteric incision **6 cm proximal and distal to the rectosigmoid junction** → antirefluxing ureteric implantation → reconfigure and close → **low-pressure rectosigmoid reservoir**. Used in exstrophy; continence good in selected patients; **acidosis still significant (whole colon exposed); adenocarcinoma risk unaltered**; commonest early complication **ureterocolonic stricture**; upper tract protection unproven long term (Chapter 134).
- **Kock colorectal valve: intussuscepted nipple valve stabilised with permanent staples** confines urine distally; distal rectum opened and patched with ileum → low reservoir pressure; no alkali therapy needed at short follow-up; long-term paediatric data awaited.
- **Tumour remains the last major concern:** a Boston series of children after ureterosigmoidostomy included adenocarcinomas with deaths.
- **Hemi-Kock augmentation of the distal rectum** with a colorectal valve: the **afferent nipple keeps stool away from the ureteroileal anastomoses**, perhaps lowering tumour risk; complex, with high early and late complications, though children were thought better suited. Others proposed the rectum augmented with a **nonrefluxing ileocaecal conduit**.
- **Prerequisite for any variant: a competent anal sphincter** — manometry, EMG and practical **retention of an oatmeal enema upright** without soiling. **Neurogenic patients who cannot hold diarrhoea are not candidates.** Stream-separating procedures within the rectal sphincter are rarely used in children.

URETEROSIGMOIDOSTOMY AND ITS VARIANTS

Variant	Key construct	Acidosis	Tumour risk
Classic	Ureters into intact sigmoid	Significant	High , long latency
Mainz II (sigma rectum)	Detubularised rectosigmoid, low pressure	Still significant	Unaltered
Kock colorectal valve	Stapled intussuscepted valve + ileal patch	Avoided (short term)	Remains
Hemi-Kock rectal augmentation	Afferent nipple isolates ureters from stool	Reduced	Possibly lower

- **Pattern:** each modification solves pressure, then acid, then stool contact in turn — and tumour is the last problem standing. **Trap:** Mainz II lowers pressure but does not remove acidosis or cancer risk.

Nipple Valves

- **Largest experience: Kock pouch;** Skinner's modifications maintain the efferent nipple, yet **failure of 15% or more** is expected and **reoperation of about one-third**, mostly for the efferent nipple; equivalent results in children, with acidosis and new hydronephrosis attributed to patient complexity.
- **Mainz I pouch:** nipple evolved — intussuscepted ileum stapled, passed through the intact ileocaecal valve and fixed again; **low incontinence with the latest modifications**, good upper tracts, rare metabolic problems in children.
- **Principle** (Figure 48.16A): **the same pressures that create continence tend to efface nipple valves.**

Flap Valves and the Mitrofanoff Principle

- **Mitrofanoff (1980):** appendix (and ureter) as a flap valve — **any tubular structure can be implanted effectively into a low-pressure reservoir**; avoids harvesting the ileocaecal valve or other segments.
- **Mechanism:** submucosal tunnel for a **supple, small-diameter conduit**; as the reservoir fills, **rising pressure transmitted through the epithelium coapts the conduit** (Figure 48.16B).
- **Appendix is ideal:** removed without significant morbidity; small calibre permits a short tunnel — **continence with only a 2-cm appendiceal tunnel** (Kaefer and Retik, 1997); relatively long in children with a thinner abdominal wall.
- **The flap valve is probably the most reliable surgical continence mechanism** — some never leak per stoma → **no pop-off** → **upper tract deterioration or rupture if catheterisation lapses.**
- **Appendix in situ** → **the reservoir must include right colon**; the **mesoappendix usually allows mobilisation** to native bladder or almost any reservoir.

Technique (appendicovesicostomy)

Low midline or transverse incision; if the caecum is high, mobilise ascending colon along the line of Toldt (laparoscopic assistance possible) → amputate the appendix base with a **small caecal cuff** (cuff at the stoma may reduce stenosis); close caecum as for appendicectomy; **harvest a flap of caecum in continuity and tubularise it if appendix length is marginal** → choose implantation site by appendix length, bladder mobility and stoma site; tunnel the distal end **posterolaterally** (anterior if anatomy suits), bladder opened or not (Figure 48.17); an **extravesical trough** covered by laterally mobilised serosa also works → bring the base (caecal cuff) to the **umbilicus or right lower quadrant without tension, without twisting or fascial compression of the pedicle**; **umbilicus can hide the stoma** → **skin flaps against stenosis** (a rotational umbilical flap helps when the mesentery is short) → **keep the conduit as short as possible**; fix appendix and bladder to peritoneum beneath fascia to prevent kinking with filling → **catheterise after each step; if not easy, revise the previous step**; also at varying volumes. Same principles apply laparoscopically and robotically.

- **Split appendix** for both caecostomy and catheterisable channel: reliable in selected patients with meticulous technique; no difference in stomal revision up to 15 years.

- **Stent 6–12 Fr for 10–14 days → surgeon personally performs the first catheterisation → CIC at least every 4 hours** by day for drainage, patency and prevention of stenosis.
- **Appendix unavailable:** prior appendicectomy, position or length, congenital absence, adhesions, or **reserved for antegrade continence enemas**; histological abnormalities are common and increase with age but **rarely preclude use**.

Alternatives

- **Ureter (Mitrofanoff): preserve distal blood supply**; refluxing ureters usable after extravesical reimplantation; **more stomal stenosis than appendix** (blood supply); catheter distension causes **discomfort**.
- **Fallopian tube** (Woodhouse and MacNeily, 1994): catheterisable, but **stenosis significant**; **consider fertility** when the ipsilateral gonad is normal.
- **Tapered ileum** (Adams et al., 1992): narrowed along the mesenteric border with **permanent staplers** in series; easy catheterisation, good continence; long enough to reach skin without tension but **short and straight** — an unnecessarily long, mobile channel kinks, causing difficult catheterisation or perforation.
- **Yang-Monti** (Yang, 1993; Monti et al., 1997): **1–2 cm of ileum**, harvested independently or adjacent to the augmentation segment, **opened along the antimesenteric border, closed transversely → circumference becomes length**; small central mesentery; mesentery-free ends ease tunnelling and exit; an **off-centre incision makes one limb longer** — short limb in the bladder, long limb through a thick abdominal wall; long intravesical tunnel unnecessary; tubularised over **12 Fr**, two layers, absorbable (Figure 48.18).
 - **Efficient use of bowel**; possibly **easier to catheterise** (mucosal folds redirected longitudinally); ileal stomas may have **less stenosis than appendix**.
 - **Disadvantage: short** — tension in obese patients despite skin flaps → stenosis → **double Monti** (two channels anastomosed) or **spiral Monti**; similar continence, stenosis and revision, but **spiral Monti more often needs subfascial revision at the umbilicus**.
 - **Pouch-like dilatation** causing difficult catheterisation is largely avoidable; worsened by **excessive extravesical length**.

Results

- **Continence superb — usually above 95%**; inability to use the appendix is rare (other than for antegrade enemas).
- **Incontinence is rare → inadequate flap-valve length or persistently high reservoir pressure → urodynamics**.
- **Stomal stenosis is the commonest complication**; also false passage; difficulty may reflect **redundant extravesical length**; early stenosis may need formal revision; **"L stent" with topical steroid cream** is a noninvasive option.
- **Suprafascial or subfascial revision is common**; **Monti channels are about twice as likely as appendicovesicostomy to need revision**; age-independent, clustering in the first months to years.
- **Benign fibroepithelial polyps and inflammatory granulation tissue** occur in all channel types; many symptomatic, often recurrent after resection.
- Rare stricture and necrosis, particularly of **caecal extensions**; abdominal stomas carry a **higher risk of reservoir calculi** (incomplete emptying).

CATHETERISABLE CHANNELS COMPARED

Channel	Advantages	Disadvantages
Appendix (Mitrofanoff)	First choice ; 2-cm tunnel suffices; minimal harvest morbidity	Stomal stenosis (umbilicus); may be needed for enemas

Channel	Advantages	Disadvantages
Yang-Monti	1–2 cm ileum; efficient; easy catheterisation	Short (obesity); twice the revisions ; pouch-like dilatation
Tapered ileum	Adequate length	Stapled; kinks if long
Ureter	Urothelial	More stenosis ; discomfort; distal blood supply
Fallopian tube	Catheterisable	Stenosis; fertility
Ileocaecal valve (Indiana)	Simple; short learning curve	Neurogenic bowel diarrhoea
Casale vesicostomy	Extraperitoneal; no bowel; appendix spared	High stenosis ; consumes bladder

- **Pattern:** channels differ chiefly in stenosis, length and what they cost elsewhere. **Trap:** stomal stenosis, not incontinence, is the commonest channel problem.

Ileocecal Valve

- **Gilchrist et al. (1950)**, popularised by the **Indiana group**: short terminal ileum, imbricated or tailored, as efferent limb — **short and straight** for easy catheterisation.
- **Continence depends on the imbricated ileocaecal valve, not the length of the efferent limb**; imbrication with **interrupted permanent sutures** across distal ileum and valve, carried onto the caecum.
- **Indiana pouch in children as good as in adults**; after the appendix, **the simplest mechanism with the shortest learning curve**; high continence with preserved upper tracts (one report less favourable).
- Caecal augmentation with the efferent limb as continent stoma: good results, little gastrointestinal effect in selected neurogenic patients.

Continent Vesicostomy

- **Yachia (1997)**: anterior bladder wall tube **woven through rectus muscle** for compression; complete continence in a small series, not duplicated.
- **Hanna et al. (1999)**: bladder or intestinal flap tubularised over **14–16 Fr**, bladder plicated around the proximal 3 cm with nonabsorbable suture — a **fundoplication-like nipple**; **Macedo and Srougi (2000)**: similar at initial augmentation — appealing without an appendix, but **nipple valves are hard to keep fixed: the forces that create continence efface the mechanism**.
- **Casale (1991)**: flap valve from a tubularised strip of bladder mucosa; suits a **compliant, large-capacity bladder**.

Technique

Parallel incisions 3 cm apart in the anterior bladder create a long rectangular full-thickness flap, measured against the abdominal wall to reach skin without tension → tubularised to the bladder in **two layers**; muscle kept broad for blood supply; mucosa trimmed to avoid redundancy → an in-line **intravesical mucosal strip 2–3 cm long, 1.5 cm wide** (Figure 48.19: further 2.5 cm) is incised and its edges mobilised — preferably one edge across to the other, avoiding overlapping suture lines — and tubularised; Casale incised the strip end transversely; a later modification **leaves it intact** → bladder mucosa on either side mobilised and closed over the mucosal tube, forming the flap valve; wider mobilisation of the opposite side avoids overlapping lines (fistula, incontinence) → bladder closure incorporates the tube to prevent kinking → **stent for 3 weeks**.

- **Strictures if not catheterised regularly — more so than other channels.**

Results

- **Continence good**, as with most flap valves; **stomal stenosis a significant problem** (high at Indiana); skin flaps and absence of tension reduce but do not eliminate it.

- **Advantages:** extraperitoneal, no bowel anastomosis, appendix reserved for enemas. **Disadvantage:** uses bladder and reduces capacity — inappropriate for many patients.

NIPPLE VALVE VERSUS FLAP VALVE

	Nipple valve	Flap valve
Examples	Kock, Mainz I, Hanna, Macedo	Mitrofanoff, Yang-Monti, Casale, Kropp
Effect of rising pressure	Effaces the valve	Coapts the conduit
Fixation	Staples; failure-prone	Submucosal tunnel
Learning curve	Longest	Short (appendix)
Continence	About 85% (stapled, experienced)	>90–95%
Hazard	Incontinence, reoperation	No pop-off if CIC fails

Results With Pediatric Continent Diversion

- **The hardest part is the efferent limb:** reliable continence plus easy catheterisation.
- **Flap valve most familiar to paediatric urologists:** appendix simple, suits most, very good continence; tapered intestinal segments when the appendix is absent or used for enemas.
- **Nipple valves: most complex, longest learning curve;** continence approaching 85% with stapled nipples after extensive experience, versus above 90% approaching 95% with other efferent limbs.
- **Hydronephrosis is rare and not increased versus conduit diversion** with good selection and technique — but **new hydronephrosis when catheterisation is unreliable.**
- **Complications will rise with follow-up:** the same as augmentation — infection, hydronephrosis, calculi, spontaneous perforation, tumour — **a function of intestine as reservoir; more bowel than augmentation → possibly more complications;** raised chloride, lowered bicarbonate and acidosis already noted; spontaneous perforation reported but rare.
- **Commonest complication: stomal stenosis, more at the umbilicus with appendix than with tapered ileum;** skin flaps into the channel lower but do not eliminate it.
- **No efferent limb is superior by consensus → individualise.**

KEY POINTS: CONTINENT URINARY DIVERSION

- "Pure" continent diversions have been used in children with results similar to adults.
- Flap valves provide a reliable continence mechanism with appendix or reconfigured intestinal segments.
- The efferent limb should always be kept as short and straight as possible to facilitate catheterisation.
- Stomal stenosis is the commonest complication of paediatric continent diversion.

Other Considerations

Use of Buttons as Alternative Means to Empty Bladder

- **Mic-Key button** (Avanos Medical) and **Mini balloon button** (Applied Medical Technology): bladder drainage only — intermittent or continuous depending on the drainage adaptor.
- **Ideal patient: not a candidate for augmentation and/or CIC.**

Technique

- Can be placed percutaneously under cystoscopic guidance, but the authors **first place a suprapubic tube to establish a tract** → button placed as an outpatient, with or without a guidewire; sizing tool supplied; **length changed with body habitus** over time.

Transitional Care Considerations

- **Transitional urologist**: emerging field for adults with paediatric reconstructions (Chapter 65).
- **Mortality from medical noncompliance** (Husmann, 2017, augmented cohort): **spontaneous rupture**, with a death; contributors **substance abuse, mental or physical deficits, catheterisation noncompliance**.
- **240 mL of saline irrigation daily was highly effective in reducing calculi** (Husmann, 2016).
- **Augmentation with concomitant bladder neck procedure** → **new renal scarring and CKD stage 3 or higher** in a substantial proportion at long follow-up, driven by **catheterisation noncompliance**.
- **Bladder cancer in spina bifida: about 2–4% overall**; most presented with **stage III or IV disease**; **non-gastric augmentation did not appear to increase risk** → counsel all patients.
- **Lifelong risk**: adult providers must assess **catheterisation compliance, daily irrigation and upper tract deterioration**, particularly after augmentation with bladder outlet procedures.

Concept — *continence mechanisms divide by **what reservoir pressure does to them**: pressure seals a flap valve tighter and effaces a nipple, so flap valves dominate paediatric practice — and their hazard is the mirror image, no pop-off when catheterisation fails.*

SUMMARY

- **Treat bladder dysfunction early and aggressively**, whatever the cause, to minimise the number needing reconstruction.
- **Evaluate carefully so every problem is identified and addressed**; be **flexible** — **fit segment and technique to the patient**; **never force one procedure on every patient**.
- **Identify and correct upper tract obstruction or VUR** at surgery; **low-grade secondary reflux usually resolves** once bladder dysfunction is corrected.
- **Provide an adequate reservoir** holding, at low pressure, the volume produced between voids or catheterisations — by augmentation or a continent reservoir, with any gastrointestinal segment.
- **Create outflow resistance at the bladder neck** if it is lacking.
- **Every patient must be prepared and able to perform CIC reliably**; most need it routinely, particularly neurogenic patients.
- **Preserve as much native urinary tract as possible** — urothelium avoids intestinal morbidity; bowel is not a perfect physiologic substitute.
- **Lifelong follow-up**, including true quality of life; **motivation of patient and family is the most important factor**, and its assessment is critical.